

100 Days of Azure DevOps

Day 45 Handout — Modules are boundaries. what-if is the dress rehearsal. A Delete you did not expect is an incident that has not happened yet

5 - Infrastructure as Code · Personal lab only · Educational content · Not a sales pitch

Architecture A — composition vs a 2,000-line main.bicep

Object	File	Call site
Module	infra/modules/storage.bicep	params: prefix, sku, location
App	infra/main.bicep	module stg 'modules/storage.bicep' = { name: 'storage', params: { ... } }
what-if	CLI against main.bicep	az deployment group what-if — Create/Ignore/Modify/Delete
Deployment stacks	Literacy: track + prune	Not required for the lab apply
ARM still underneath	Compiled JSON per module	Errors still speak resource IDs

Architecture B — when to stop at what-if

what-if result	Action	Why
Create (expected storage)	Deploy	Matches the file
Modify SKU	Read twice, then deploy if you meant it	SKU change can recreate
Delete you did not type	Stop	Wrong RG, Complete-mode surprise, or name change = replace
NoChange / Ignore	Good	Idempotency passing the adult test

Architecture C — pitfalls

Pitfall	What you see	Fix
Name change in module	Delete+Create of the storage account	Treat name as identity; what-if before apply
Skipped what-if	'Only a SKU change' recreates a disk	Make what-if a required step, including in CI later
Module path typo	Build fails / empty deploy	Relative path from main.bicep
what-if unread in CI	Log has a Delete, human did not look	Eyes on Modify/Delete. Automation without eyes is a faster rumor

One-liner to remember

what-if is a dress rehearsal — read the diff before the audience (prod) arrives.

Step-by-step lab (personal Azure / Azure DevOps only)

Personal subscription. RG rg-day45-lab. Terraform-primary: still run one what-if so the verb exists in your hands.

- 1 Split storage into infra/modules/storage.bicep. main.bicep only calls the module.
- 2 az group create -n rg-day45-lab -l centralindia.
- 3 az deployment group what-if -g rg-day45-lab -f infra/main.bicep. Read every Delete/Modify line out loud.
- 4 If what-if matches intent: az deployment group create -g rg-day45-lab -f infra/main.bicep. Run what-if again — expect NoChange.
- 5 If a Delete appears that you did not expect: stop. Do not 'just apply'.

6 az group delete -n rg-day45-lab --yes --no-wait.

Module call + what-if (required manners)

```
// infra/main.bicep
param location string = resourceGroup().location
module stg 'modules/storage.bicep' = {
  name: 'storage'
  params: {
    location: location
    skuName: 'Standard_LRS'
  }
}

az deployment group what-if -g rg-day45-lab -f infra/main.bicep
# Read Create / Modify / Delete. Then, only if boring:
az deployment group create -g rg-day45-lab -f infra/main.bicep
```

Done checklist

- Storage lives in a module, not an inlined blob forever
- what-if run before create
- Second what-if is idle (idempotent) or you stopped on surprise Delete
- RG deleted

Tomorrow — Day 46

Terraform basics — init, plan, apply, destroy. State is memory; lose it and you argue with ghosts.